REVISION 2021 • SEMESTER 2 • ENGINEERING SCIENCE

Engineering Mechanics

A complete diploma engineering handbook for Course Code 2021. It converts the official syllabus into a practical learning and exam-preparation text covering force systems, equilibrium, beams, trusses, friction, centroid, centre of gravity, moment of inertia, stress, strain, elastic constants, solved numerical problems, practice questions and a fresh model question paper.

COURSE CODE
2021

COURSE TITLE
Engineering Mechanics

SEMESTER
2

CREDITS
3

CATEGORY
Engineering Science

PERIODS/WEEK
3 (L:2 T:1 P:0)

PERIODS/SEMESTER
45

PROGRAMME
Architecture, Automobile, Civil allied, Mechanical allied, Wood & Paper Technology, Environmental Engineering

Course overview and official source data

Official information is taken from the supplied syllabus PDF. Missing official items are not guessed.

What the subject is

Engineering Mechanics studies the effect of forces on bodies. In this course the focus is mainly statics and basic strength: how to draw a free body diagram, how to find support reactions, how to calculate centroid and moment of inertia, and how to predict stress and strain in materials.

□□□□□□□□□□□: load □□□□□□ body □□□□□□ balance □□□□□□□□□□□, support reaction □□□□□□□□□, material safe □□□ □□□□□□ □□□□□□□□□□□□□□ subject □□□ Engineering Mechanics. Formula □□□□□□ □□□□; FBD □□□□□□□ □□□□□□□.

Industrial relevance

- ✓ Beam and bracket reaction checking.
- ✓ Machine frame and panel support design.
- ✓ Truss, roof and escalator structural load-path thinking.
- ✓ Friction in clamps, brakes, rollers and moving parts.
- ✓ Stress-strain and material testing.

Confirmed official information

Item	Value
Programme / branch	Diploma in Architecture / Automobile Engineering / Civil Engineering & allied programs / Mechanical Engineering & allied programs / Wood & Paper Technology / Environmental Engineering
Course code	2021
Course title	Engineering Mechanics
Semester	2
Credits	3
Course category	Engineering Science
Periods per week	3 (L:2 T:1 P:0)
Periods per semester	45
Objective	To impart knowledge and skills that enable the student to predict the effects of force and motion.

Prerequisites	Integration and trigonometry from Mathematics I & II; scalar/vector quantities and triangular/parallelogram law of forces from Applied Physics I.
Suggested practicals	Not specified in supplied syllabus/source.
Official model QP pattern	Not specified in supplied syllabus/source. A fresh practice paper is included here for preparation.

Course outcomes

CO	Outcome	Engineering meaning	Hours
CO1	Identify force systems using basics of mechanics.	Classify forces, draw FBD, resolve forces, find resultant, moment and equilibrant.	10
CO2	Apply static equilibrium to determine unknown forces.	Find beam reactions, simple truss forces and friction quantities.	12
CO3	Solve rigid body problems using properties of distributed areas/masses.	Find centroid, CG, M.I., radius of gyration and polar M.I.	11
CO4	Determine structural behaviour of materials under loading.	Calculate stress, strain, modulus and interpret stress-strain curve.	10

CO-PO mapping

CO1 → PO1 moderately. CO2 → PO1 strongly and PO2 moderately. CO3 → PO1 strongly. CO4 → PO1 strongly.

How to study

1. Draw body/load diagram and FBD first.
2. Write known data with units.
3. Choose equations: $\Sigma Fx = 0$, $\Sigma Fy = 0$, $\Sigma M = 0$.
4. For centroid/M.I., use table method.
5. For stress-strain, check units carefully.

Redesigned syllabus roadmap

A practical route showing what to draw, calculate, explain and avoid.

Module	CO	Hours	Core topics	Draw	Calculate	Application	Common mistakes
M1 Basics of mechanics	CO1	10	Mechanics, statics, dynamics, scalar/vector, SI units, resolution, resultant, moment, Varignon, FBD, Lami.	Force components, FBD, Lami triangle.	Resultant, moment, equilibrant, string tensions.	Brackets, lifting points, machine supports.	Wrong angle, no FBD, no direction in answer.
M2 Beams, truss and friction	CO2	12	Beam types, supports, loads, reactions, simple truss, method of sections, friction.	Beam loading, truss joint, friction block.	Support reactions, truss member forces, friction and angle of repose.	Frames, roof truss, rollers, brakes, clamps.	UDL resultant at wrong point, wrong friction direction.
M3 Centroid, CG and MI	CO3	11	Centroid of simple/composite figures, CG of solids, M.I., radius of gyration, theorems, polar M.I.	Composite section axes, I/T/channel/angle sections.	Centroid, M.I., radius of gyration, polar M.I.	Section selection for beams and frames.	Forgetting negative area and Ad^2 term.
M4 Stress and strain	CO4	10	Rigid/elastic/plastic bodies, stress, strain, Hooke, stress types, stress-strain curve, properties, elastic constants.	Stress-strain curve, axial load sketch.	Stress, strain, elongation, E, G, K, Poisson ratio.	Material testing, rods, bolts, shafts, supports.	Area unit error, strain with unit, stress/load confusion.

Module 1 • CO1 • 10 hours

Basics of mechanics and force systems

Identify force systems, resolve forces, determine resultant, compute moment, draw free body diagrams and solve simple Lami theorem problems.

Mechanics and force systems

Mechanics is the science of force and motion. Applied mechanics uses mechanics to solve engineering problems. Statics deals with bodies in equilibrium, while dynamics deals with bodies in motion. A force system may be concurrent, non-concurrent, parallel, non-parallel, coplanar or non-coplanar.

Rigid body

Deformation is neglected. Beam and truss reactions usually use this assumption.

Particle

Size is ignored; useful for force equilibrium at a joint.

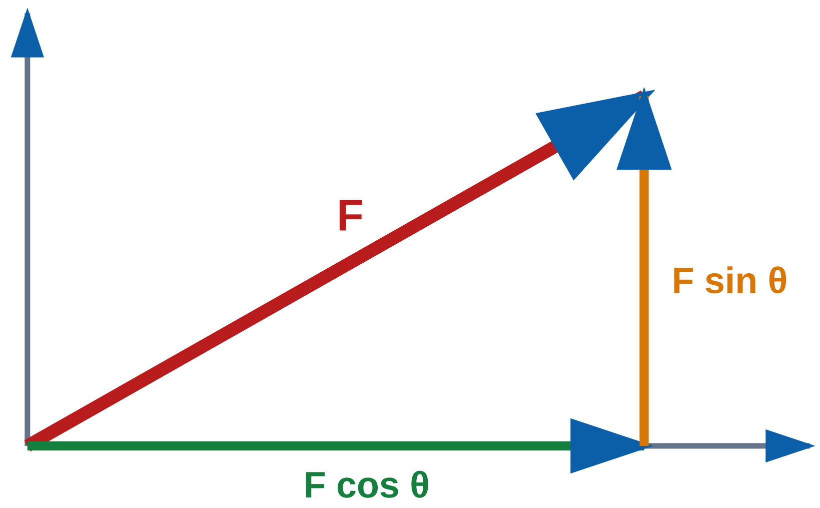
Scalar

Magnitude only: mass, time, area.

Vector

Magnitude and direction: force, velocity, moment.

Statics-□ body move □□□□□□□□□□□□□□□□□ □□□□□□ force balance □□□. ΣF = 0, ΣM = 0 □□□□□□□□□ core idea.



Resolution of an inclined force.

Resolution, resultant and moment

$F_x = F \cos \theta$, $F_y = F \sin \theta$

$$R = \sqrt{(\Sigma Fx)^2 + (\Sigma Fy)^2},\ \tan\theta = \Sigma Fy / \Sigma Fx$$

$$\text{Moment} = F \times \text{perpendicular distance}$$

Varignon's theorem states that the moment of a force about a point equals the sum of moments of its components about the same point. Torque is a twisting moment, common in spanners, shafts and screws.

Equilibrium, equilibrant and FBD

$$\Sigma Fx = 0,\ \Sigma Fy = 0,\ \Sigma M = 0$$

Equilibrant is equal in magnitude and opposite in direction to the resultant. A free body diagram isolates the body and shows all loads and reactions.

- ✓ Remove supports and replace by reactions.
- ✓ Show applied forces with direction.
- ✓ Mark dimensions and moment arms.
- ✓ Use clear sign convention.

Lami's theorem

For a body in equilibrium under three coplanar, concurrent and non-parallel forces:

$$F1/\sin\alpha = F2/\sin\beta = F3/\sin\gamma$$

Use Lami only for exactly three forces acting at a point. Angles are opposite to the corresponding forces.

Lami theorem-□ □□□ force-□□□□ opposite angle □□□ □□□□□□□□□□□□□□□□. □□□□□□ force □□□□□□ □□□□□□□□□ □□□□□□ □□□□□□□□□□□.

Common mistakes

- ✕ Using sin/cos with wrong angle.
- ✕ Moment arm not perpendicular.
- ✕ No direction for resultant.
- ✕ Using Lami for more than three forces.

Expected questions

1 mark Define force, moment, torque and equilibrant.

Short Differentiate scalar and vector quantities.

Short State principle of transmissibility.

Numerical Find resultant of concurrent forces.

Diagram Draw FBD of a loaded body.

3 mark State and explain Varignon's theorem.

Numerical Calculate moment of a force.

Numerical Solve a three-force equilibrium problem using Lami.

Module 2 • CO2 • 12 hours

Beams, supports, simple truss and friction

Calculate reactions for simple beams, understand truss member forces and solve friction problems.

Beams, supports and loads

A beam carries transverse loads. The syllabus focuses on simply supported and cantilever beams without overhang. Loads may be vertical/inclined point load, uniformly distributed load and couple.

- ✓ Roller support gives one reaction.
- ✓ Hinged support gives horizontal and vertical reactions.
- ✓ Fixed support gives force reactions and fixing moment.

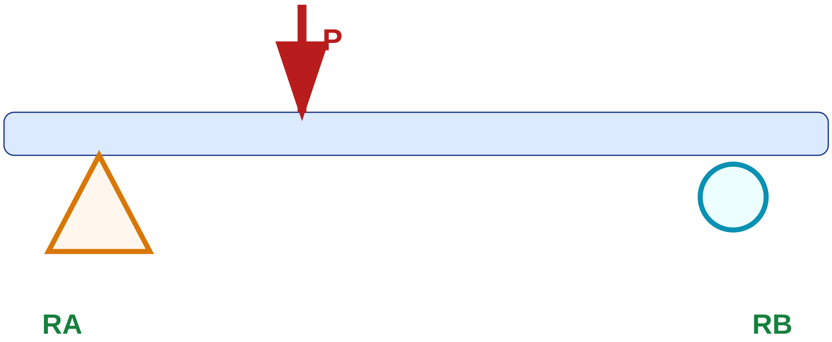
Beam reaction method

Replace UDL by equivalent point load at centroid of loaded length.

UDL resultant $W = wL$, acting at $L/2$ from UDL start

$\Sigma V = 0, \Sigma M = 0$

For a cantilever with end point load P over length L: reaction R = P and fixing moment M = PL.



Simply supported beam with point load.

Simple truss

A truss is a framework of straight members connected at joints. Loads are assumed at joints; members carry axial tension or compression. In method of joints, isolate one joint and apply $\Sigma Fx = 0$ and $\Sigma Fy = 0$.

Method of sections cuts through selected members and applies equilibrium to one part, faster when only selected member forces are required.

Friction

Friction resists relative motion or impending motion between two surfaces.

$F \leq \mu R$; limiting friction $F_{max} = \mu R$

$\tan \varphi = \mu$; $\tan \theta = \mu$ for angle of repose

Friction force □□□□□□□□□ impending/actual motion-□□ opposite direction-□ □□□. Direction □□□□□□□□□□ equation □□□□□□□□.

Field relevance

- ✓ Panel frame support behaves like a beam.
- ✓ Escalator truss members carry tension/compression.
- ✓ Friction appears in brakes, clamps and moving parts.
- ✓ Support symbols simplify real structures into calculable models.

Common mistakes

- ✗ UDL resultant placed at wrong location.
- ✗ Moment equation sign error.
- ✗ Forgetting pin horizontal reaction.
- ✗ Assuming every truss member is tension.
- ✗ Using F = μR when not in limiting condition.

Expected questions

- Name beam/support/load types.
- Draw support symbols and reactions.
- Find reactions for point load beam.
- Find reactions for UDL beam.
- Explain method of joints.
- Outline method of sections.
- State laws of friction.
- Find limiting friction and angle of friction.
- Explain angle of repose.
- Draw friction block FBD.

Module 3 • CO3 • 11 hours

Centroid, centre of gravity and moment of inertia

Locate centroid/CG and calculate moment of inertia for common and composite sections.

Centroid and centre of gravity

Centroid is the geometrical centre of an area. Centre of gravity is the point through which the total weight of a body acts. For uniform plane lamina, both coincide.

$$\bar{x} = \Sigma(Ax)/\Sigma A, \quad \bar{y} = \Sigma(Ay)/\Sigma A$$

For solids, use volume or weight instead of area.

$$\bar{x} = \Sigma(Vx)/\Sigma V, \quad \bar{y} = \Sigma(Vy)/\Sigma V$$

Basic centroid locations

Shape	Centroid
Rectangle	b/2, h/2 from corner
Triangle	h/3 from base
Circle	At centre
Semicircle	$4r/(3\pi)$ from diameter
Quarter circle	$4r/(3\pi)$ from both straight sides

Composite figures and holes

1. Choose reference axes.
2. Split section into simple shapes.
3. Prepare table A, x, y, Ax, Ay.
4. Use negative area for cut-outs.
5. Compute $\bar{x}$ and $\bar{y}$.

Composite section-□ table method □□□□□□□□□□□. Hole/cut-out area negative □□□□□□.

Moment of inertia

M.I. of area measures how far area is distributed from an axis. A larger M.I. means better resistance to bending about that axis.

$$\text{Radius of gyration: } k = \sqrt{I/A}$$

$$\text{Parallel-axis theorem: } I = I_G + Ad^2$$

Perpendicular-axis theorem: $J = I_x + I_y$

Common M.I. formulae

Section	M.I.
Rectangle b×h	$I_{xx} = bh^3/12$, $I_{yy} = hb^3/12$
Square side a	$I = a^4/12$
Circle diameter D	$I = \pi D^4/64$
Triangle	I about centroidal axis parallel to base = $bh^3/36$
Solid circular polar M.I.	$J = \pi D^4/32$

Industrial application

- ✓ I-section resists bending efficiently because area is far from neutral axis.
- ✓ Channel and angle sections are common in frames and brackets.
- ✓ Hollow sections improve stiffness-to-weight ratio.
- ✓ Polar M.I. is important for shafts.

Mistakes to avoid

- ✕ Wrong reference axes.
- ✕ Using triangle centroid from wrong side.
- ✕ Forgetting negative area for holes.
- ✕ Using $bh^3/12$ about base.
- ✕ Missing Ad^2.

Expected questions

Define centroid, CG, M.I. and radius of gyration.

State parallel-axis theorem.

Find centroid of L/T/I section.

Find centroid with cut-out hole.

Write M.I. formulas.

Find M.I. of composite section.

Explain polar M.I.

Why is I-section strong in bending?

Module 4 • CO4 • 10 hours

Simple stresses and strains

Understand stress, strain, Hooke's law, material properties and elastic constants.

Rigid, elastic and plastic bodies

- ✓ Rigid body: deformation neglected.
- ✓ Elastic body: returns to original shape within elastic limit.
- ✓ Plastic body: retains permanent deformation.

Real materials deform under load; safe design keeps stress below allowable limit.

Stress, strain and Hooke's law

Stress: $\sigma = P/A$

Strain: $\varepsilon = \delta L/L$

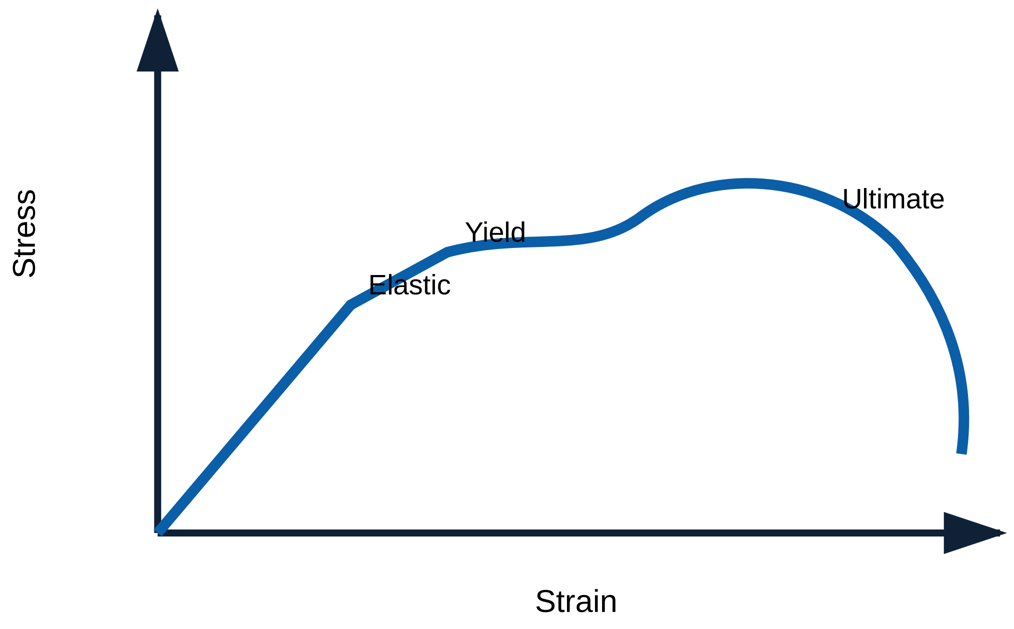
$E = \sigma/\varepsilon$ within elastic limit

Stress load □□□□; load/area □□□. Strain-□□ unit □□□□.

Types of stresses

Type	Nature	Example
Direct tensile	Pulling normal to area	Tie rod, cable

Direct compressive	Pushing normal to area	Column, strut
Shear	Force parallel to area	Rivet, pin
Bending	Due to bending moment	Beam
Torsion	Due to twisting moment	Shaft



Mild steel stress-strain curve.

Mechanical properties

Elasticity, stiffness, plasticity, toughness, brittleness, ductility, malleability and hardness describe how material responds to load, deformation, impact, forming and wear.

Elastic constants

E = longitudinal stress / longitudinal strain

G = shear stress / shear strain

K = volumetric stress / volumetric strain

Poisson's ratio μ = lateral strain / longitudinal strain

$E = 2G(1+\mu)$, $E = 3K(1-2\mu)$

Testing and safety

- ✓ Measure specimen diameter and gauge length before loading.
- ✓ Record yield, ultimate and fracture loads separately.
- ✓ Keep hands away from specimen during tensile test.
- ✓ Use factor of safety in design.

Common mistakes

- ✗ Stress without area.
- ✗ Mixing mm^2 and m^2.
- ✗ Writing unit for strain.
- ✗ Confusing elastic modulus and elastic limit.

Expected questions

Define stress, strain and Hooke's law.

Draw stress-strain curve of mild steel.

Explain yield stress and ultimate stress.

Calculate stress and strain.

Define ductility, malleability, toughness.

Calculate Poisson's ratio.

Use E, G, K relations.

Explain factor of safety.

Formula / rule bank

Variables, units and applications for quick revision.

Formula	Variables	Unit	Use	Example
$F_x=F\cos\theta$, $F_y=F\sin\theta$	F force, θ angle with x-axis	N	Resolution	100 N at 30°: Fx=86.6 N, Fy=50 N
$R=\sqrt{[(\Sigma F_x)^2+(\Sigma F_y)^2]}$	Component sums	N	Resultant	30-40 gives 50 N
$M=Fd$	d perpendicular distance	N·m	Moment	200 N at .5 m = 100 N·m
$\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$	Equilibrium equations	-	Statics	Beam reactions
$F1/\sin\alpha=F2/\sin\beta=F3/\sin\gamma$	Three concurrent forces	N	Lami	Suspended load
$W=wL$	w UDL, L length	kN	Beam UDL	3 kN/m × 2 m = 6 kN
$F=\mu R$	μ coefficient, R normal reaction	N	Limiting friction	μ=.25, R=800 N, F=200 N
$\bar{x}=\Sigma Ax/\Sigma A$	A area, x coordinate	mm	Centroid	L-section table
$I=IG+Ad^2$	d axis distance	mm^4	Parallel axis	Composite M.I.
$Ixx=bh^3/12$	b breadth, h depth	mm^4	Rectangle M.I.	Beam section
$\sigma=P/A$	P load, A area	N/mm^2	Stress	10 kN/100 mm^2 = 100 N/mm^2
$\varepsilon=\delta L/L$	δL extension, L original length	No unit	Strain	1/500=.002
$E=\sigma/\varepsilon$	Young's modulus	N/mm^2	Elasticity	Stress/strain
$E=2G(1+\mu)$	G rigidity modulus	N/mm^2	Elastic constants	Find G

Solved examples / problem lab

Show formula, substitution, calculation and final answer with unit.

1. Resultant

Forces 30 N east and 40 N north.

$$R=\sqrt{(30^2+40^2)}=50\text{ N}$$

$$\tan\theta=40/30;\ \theta=53.13°$$

Answer: 50 N at 53.13° north of east.

2. Moment

250 N acts at 0.32 m perpendicular distance.

$$M = Fd = 250 \times 0.32 = 80\ \mathrm{N \cdot m}$$

Answer: 80 N·m.

3. Beam point load

Span 6 m, load 12 kN at 2 m from A.

$$RA + RB = 12$$

$$RB \times 6 = 12 \times 2;\ RB = 4\ \mathrm{kN}$$

RA=8 kN. Answer: RA=8 kN, RB=4 kN upward.

4. Beam UDL

Span 5 m, UDL 4 kN/m over full span.

$$W = wL = 4 \times 5 = 20\ \mathrm{kN\ at\ centre}$$

Symmetry: RA=RB=10 kN.

5. Friction

Block W=800 N, μ=.25.

$R=800\text{ N},\ F=\mu R=.25\times800=200\text{ N}$

$\varphi=\tan^{-1}(.25)=14.04°$

Answer: F=200 N, φ=14.04°.

6. Centroid of L-section

Vertical rectangle 100×20, horizontal 80×20, overlap 20×20.

A1=2000 at (10,50), A2=1600 at (40,10), A3=-400 at (10,10). ΣA=3200.

$\bar{x}=(2000\times10+1600\times40-400\times10)/3200=25\text{ mm}$

$\bar{y}=(2000\times50+1600\times10-400\times10)/3200=35\text{ mm}$

7. M.I. of rectangle

Rectangle 60 mm × 100 mm about centroidal x-axis.

$I_{xx}=bh^3/12=60\times100^3/12=5.0\times10^6\text{ mm}^4$

8. Stress and strain

Rod diameter 20 mm, load 30 kN, length 1 m, extension .095 mm.

$A = \pi d^2/4 = 314.16\ mm^2$

$\sigma = 30000/314.16 = 95.49\ N/mm^2$

$\varepsilon = .095/1000 = 0.000095$

$E = \sigma/\varepsilon = 1.005 \times 10^6\ N/mm^2$

9. Elastic constant

E=200 GPa, μ=.25.

$G = E/[2(1+\mu)] = 200/[2 \times 1.25] = 80\ GPa$

10. Lami theorem

For a 600 N load held by strings, opposite angles are 150°, 135° and 75°.

$T1/\sin 150 = T2/\sin 135 = 600/\sin 75$

T1=310.6 N, T2=439.2 N.

Practice and quick revision

Quick revision

- ✓ Force is vector.
- ✓ Moment needs perpendicular distance.
- ✓ Equilibrium means force and moment balance.

- ✓ FBD starts every statics problem.
- ✓ UDL acts at centroid of loaded length.
- ✓ Truss members carry axial force.
- ✓ Friction opposes motion tendency.
- ✓ Holes use negative area.
- ✓ M.I. increases when area moves away from axis.
- ✓ Strain has no unit.

Objective questions

Vector quantity: A Mass B Time C Force D Area

Unit of moment: A N B N/m C N·m D m/N

Equilibrium equations: A ΣF only B ΣM only C ΣFx=0,ΣFy=0,ΣM=0 D None

Lami applies to: A four forces B three concurrent coplanar equilibrium forces C parallel forces D moving bodies

UDL 5 kN/m over 4 m equals: A 5 B 10 C 20 D 25 kN

Roller support gives: A two reactions B one reaction C fixed moment D no reaction

Friction acts: A along motion B normal C opposite motion tendency D vertical

Centroid of rectangle: A corner B diagonal midpoint C one-third height D outside

Area M.I. unit: A mm^2 B mm^3 C mm^4 D N/mm^2

Stress is: A load×area B load/area C area/load D extension/length

Strain unit: A N B mm C N/mm^2 D no unit

Hooke's law valid up to: A breaking point B ultimate C elastic limit D plastic range

Answer key

1-C, 2-C, 3-C, 4-B, 5-C, 6-B, 7-C, 8-B, 9-C, 10-B, 11-D, 12-C.

Module-wise practice problems

M1: Resolve 100 N at 40°.

M1: Resultant of 60 N and 80 N at right angle.

M2: Beam span 8 m carries 20 kN at 3 m from A. Find reactions.

M2: UDL 2 kN/m over 6 m. Find reactions.

M2: W=1000 N, μ=.2. Find friction and angle.

M3: Find centroid of T-section.

M3: Find Ixx of 80×120 rectangle.

M4: Area 250 mm^2 carries 50 kN. Find stress.

Fresh model question paper

Official model question paper pattern is not specified in the supplied syllabus/source. This is a fresh practice paper for full syllabus coverage.

Course Code: 2021
Course Title: Engineering Mechanics
Time: 3 Hours
Maximum Marks: 100

PART A - One mark questions (10 x 1 = 10)
1. Define force.
2. What is a free body diagram?
3. State SI unit of moment.
4. What is an equilibrant?
5. Name two beam supports.
6. Define coefficient of friction.
7. Write centroid location of rectangle.
8. Define radius of gyration.
9. Define stress.
10. What is Poisson's ratio?

PART B - Short answer questions (10 x 3 = 30)
11. Differentiate scalar and vector quantities.
12. State Varignon's theorem.
13. Explain principle of transmissibility.
14. Draw simply supported beam with point load and reactions.

15. Differentiate hinged and roller support.
16. State laws of friction.
17. Explain centroid formula for composite area.
18. State parallel-axis theorem.
19. Differentiate tensile and shear stress.
20. Draw stress-strain curve of mild steel.

PART C - Analytical / numerical questions (Answer any six, 6 x 10 = 60)
21. A 120 N force acts at 35° above horizontal. Find components.
22. Forces 50 N and 80 N act at 60°. Find resultant.
23. Simply supported beam span 7 m carries 21 kN at 3 m from A. Find reactions.
24. Beam span 6 m carries UDL 4 kN/m over whole span. Find reactions.
25. Block weight 900 N, μ=.30. Find limiting friction and angle of friction.
26. Find centroid of T-section: flange 100×20 mm, web 20×100 mm, web centered.
27. Find M.I. of rectangle 80 mm wide, 120 mm deep about centroidal x and y axes.
28. Circular steel rod 25 mm diameter carries 40 kN tensile load. Find stress.
29. A 1.5 m bar extends .75 mm under stress 150 N/mm². Find strain and E.
30. E=210 GPa and μ=.30. Find modulus of rigidity.

Complete model answers and references

Part A

1 Force changes or tends to change rest/motion/shape. 2 FBD is isolated body with all external forces and reactions. 3 N·m. 4 Equilibrant is equal and opposite to resultant. 5 Hinged, roller, fixed, simple. 6 μ = limiting friction / normal reaction. 7 At b/2, h/2. 8 $k=\sqrt{(I/A)}$. 9 $\sigma=P/A$. 10 Lateral strain/longitudinal strain.

Part B guide

11-15

Give definitions and examples for scalar/vector; state Varignon; explain force can shift along its line of action for rigid body; draw beam with RA/RB; hinged gives two reactions, roller gives one.

16-20

Friction opposes motion and $F=\mu R$ at limiting condition; centroid formula $\bar{x}=\Sigma Ax/\Sigma A$; parallel axis $I=I_G+Ad^2$; tensile stress normal pulling, shear stress tangential; stress-strain curve must show elastic, yield, ultimate and fracture regions.

21 Components

$$Fx=120\cos35=98.30\ N$$

$$Fy=120\sin35=68.83\ N$$

Angle is 35° above horizontal.

22 Resultant

$$R=\sqrt{(50^2+80^2+2\times50\times80\times\cos60)}=113.58\ N$$

23 Beam point load

$$RA+RB=21$$

$$RB\times7=21\times3 \rightarrow RB=9\ kN$$

RA=12 kN.

24 UDL beam

$$W=4\times6=24\ kN$$

Symmetry: RA=RB=12 kN.

25 Friction

$$F = .30 \times 900 = 270\ \text{N}$$

$$\varphi = \tan^{-1}(.30) = 16.70^\circ$$

26 T-section centroid

Flange A1=2000 at y=110 mm; web A2=2000 at y=50 mm.

$$\bar{y} = (2000 \times 110 + 2000 \times 50)/4000 = 80\ \text{mm}$$

Centroid lies on symmetry axis, 80 mm from bottom.

27 Rectangle M.I.

$$I_{xx} = 80 \times 120^3/12 = 11.52 \times 10^6\ \text{mm}^4$$

$$I_{yy} = 120 \times 80^3/12 = 5.12 \times 10^6\ \text{mm}^4$$

28 Stress

$$A = \pi \times 25^2/4 = 490.87\ \text{mm}^2$$

$$\sigma = 40000/490.87 = 81.49\ \text{N/mm}^2$$

29 Strain and E

$\varepsilon = .75/1500 = .0005$

$E = 150/.0005 = 300000\ N/mm^2$

30 Modulus of rigidity

$G = E/[2(1+\mu)] = 210/[2 \times 1.30] = 80.77$ GPa

References

Official references from supplied syllabus: R.S. Khurmi Applied Mechanics; R.K. Bansal Engineering Mechanics; Ramamrutham Engineering Mechanics; Bedi Strength of Materials; Khurmi Strength of Materials; Ramamurtham Strength of Materials; Punmia Strength of Materials; Ram & Chauhan Applied Mechanics; Meriam & Kraige Engineering Mechanics Statics; Rattan Strength of Materials. Official online resources listed: NPTEL courses 122104015 and 112103109.